

**INTEGRATED CLIMATE-INTELLIGENT SALT PRODUCTION
ECOSYSTEM: A MULTI-MODAL AI FRAMEWORK FOR
SUSTAINABLE YIELD OPTIMIZATION IN TROPICAL
COASTAL ENVIRONMENTS**

25-26J-431

BSc (Hons) degree in Information Technology
Specializing in Software Engineering

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DECLARATION

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ABSTRACT

The Puttalam salterns in Sri Lanka are home to the country's solar salt industry, which has long kept the country self-sufficient in salt through traditional solar evaporation. But during the 2023-2024 production cycle, there was an unexpected monsoon that brought 1,895 mm of rain, which caused production to drop by more than 40%. This forced the Ministry of Industries to bring in 30,000 metric tonnes of non-iodized salt at the start of 2025. This crisis showed that saltern operators are very vulnerable because they rely on manual logbooks, informal price negotiations, and experience-based harvest timing, and they don't have access to any data-driven decision support. This dissertation introduces BrineX, an Integrated Climate-Intelligent Salt Production Ecosystem created in partnership with the Puttalam Salt Society. It consists of four interconnected AI-powered modules: Crystal, Compass, Vision, and Valour.

Crystal, the crystallization forecasting module, uses a dual-input Bidirectional Long Short-Term Memory (LSTM) network that has been trained on 1,095 daily records of eight pond crystallization parameters and fourteen weather features from the OpenWeatherMap API. It predicts daily crystallization conditions for the next 60 days. A causally specified Linear Regression model converts these daily outputs into monthly production volume forecasts, attaining a R^2 of 0.9732 based on 36 months of PSS operational data - an enhancement of 0.5632 R^2 compared to an ARIMA ensemble baseline. Crystal sends monthly production forecasts to Compass, the market intelligence and planning module, through a gRPC-to-HTTP bridge. Compass then turns these forecasts into three types of per-landowner market intelligence: demand estimates using a seasonal yield ratio algorithm (Maha: 10.27%, Yala: 8.74%, Transition: 9.51%; $\sim 10.0\%$ MAPE on held-out PSS data), salt price forecasts using a SARIMAX(1,1,1)(0,1,1,12) model achieving 1.16% MAPE at month+1 and 1.35% MAPE at month+2 with rolling-origin cross-validation, and ranked buyer recommendations via a fully transparent rule-based ranking system — chosen over Reinforcement Learning (13.5% Top-1 accuracy) and CatBoost (structurally unnecessary complexity). Vision, the quality control module, uses a YOLOv8-medium object detection model trained on 1,952 labelled images from Puttalam salterns to automate the detection of salt impurities and the grading of quality across multiple parameters. The model was exported to ONNX for deployment that doesn't depend on runtime. The model got 99.41% precision, 99.14% recall, and 99.37% MAPE 0.5 on a test partition of 293 images that was not used for training. It made per-crystal whiteness indices from the HSV colour space and a composite quality score to help operators make decisions in real time.

The platform uses a 13 microservices, cloud-native NestJS/TypeScript and Python/FastAPI system architecture deployed through Docker Compose with MongoDB Atlas persistence, using gRPC with Protocol Buffer contracts to communicate synchronously between services and using Apache Kafka for asynchronous event processing. Stress testing of the system confirmed the system is production-grade and has reached the readiness level, BrineX represents the first published, completely integrated, data-driven decision support platform built for the Sri Lankan solar salt industry.

ACKNOWLEDGEMENT

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LIST OF ABBREVIATIONS

Table 0.1: List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
ARIMA	Autoregressive Integrated Moving Average
AWS	Amazon Web Services
BrineX	Integrated Climate-Intelligent Salt Production Ecosystem
CI/CD	Continuous Integration / Continuous Delivery
CPU	Central Processing Unit
CRUD	Create, Read, Update, Delete
gRPC	Google Remote Procedure Call
HTTP	Hypertext Transfer Protocol
ICHORA	International Conference on Humanities, Social Sciences and Organizational Research Advancement
IoT	Internet of Things
IPCC	Intergovernmental Panel on Climate Change
KPI	Key Performance Indicator
LKR	Sri Lankan Rupee
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MASE	Mean Absolute Scaled Error
ML	Machine Learning
MongoDB	MongoDB Atlas - Cloud-hosted NoSQL Database
MAPE	Mean Absolute Percentage Error
MSE	Mean Squared Error
NestJS	Node.js Framework for Scalable Server-Side Applications
ONNX	Open Neural Network Exchange
PSS	Puttalam Salt Society
R ²	Coefficient of Determination
REST	Representational State Transfer
RL	Reinforcement Learning
RMSE	Root Mean Square Error
SaaS	Software-as-a-Service
SARIMAX	Seasonal Autoregressive Integrated Moving Average with Exogenous Variables
SLSI	Sri Lanka Standards Institute
SLIIT	Sri Lanka Institute of Information Technology
SQL	Structured Query Language
UI	User Interface
VU	Virtual User

1. INTRODUCTION

1.1. BACKGROUND AND LITERATURE SURVEY

1.1.1. The Sri Lankan Solar Salt Industry

Today, the solar evaporation process, which produces sodium chloride (table salt) from seawater, is one of the oldest industrial processes still in existence and relies on the effects of the sun's heat, winds, and evaporation to form solid-crystalline sodium chloride through high temperatures, low rainfall, and wind until exceeding saturation point. Historically, this was how Sri Lanka achieved its goal of being self-sufficient in producing all the salt it required. The major salt-producing areas in Sri Lanka are in Puttalam, Hambantota, and Elephant Pass. These areas produce large quantities of salt for use within the country or for regulated exports. The largest and most complexly managed of these areas is the Puttalam salterns, which are primarily operated by the Puttalam Salt Society (PSS).

The crystallisation process is the most dependent upon the environment at any point in the production cycle. It is influenced by all the weather conditions present: temperature; humidity; wind speed; rainfall; and pond salinity together determine if one day will permit continued concentration of brine; or present an increased risk of the crystals dissolving leading to a loss of harvested material. In years with historically stable climate conditions, operators with substantial experience had sufficient information from past years to allow them to plan with a high degree of detail based upon historical records. However, the changing nature of weather in the tropical, South Asian region over the past ten years has resulted in operators being unable to make plans having even a minimal amount of reliance upon this experience-based planning approach.

In the production cycle for 2023-2024, the vulnerability of our salt supply could not have been clearer to see than with the unseasonable monsoon rain, with 1,895mm of rain in saltern districts leading to over 40% lower production levels than had been anticipated, resulting in a national salt supply emergency. Due to this, the Ministry of Industries had to import over 30,000 metric tonnes of non-iodized salt to the start of 2025, resulting in job losses, reduced ability to export, and a direct threat to both national food and industrial security [1]. The production crisis became a lot more than just the failure of one harvest, as it further exposed the complete lack of any kind of market planning infrastructure around the production of salt. From the initial producers' perspectives, there is an absolute absence of any ability to predict yield from the land; thus producing an unstable yield has no way of being stable from a price perspective as there is no method available to forecast production volumes, to know what the market price will be, or to identify a buyer with a price point in mind that is favourable to them and their land base. Producers are still relying on manual bookkeeping, informal price discovery through intermediaries, and simply guessing when to harvest — three methods of measurement that are structurally flawed in a fast-changing environment related to climate and market [7].

1.1.2. Climate Variability and Its Impact on Coastal Agricultural Industries

The situation through which Sri Lanka's agricultural landscape has suffered major changes mirrors a common trend observed in the region. Nirwansyah and Braun demonstrated (through the hydrodynamic modelling of solar salt farming in northern Java) that disruptions to either rainfall or tidal cycles, even moderate ones, can result in considerable declines in productivity which are highly relevant to the current situation in Puttalam [2]. Ahmad et al.'s recent research has confirmed that the frequency and intensity of monsoon anomalies experienced by coastal locations throughout South Asia are increasing because of anthropogenic climate change, creating complications for crop/livestock producers and industries reliant on stable climatic conditions [3]. The IPCC 2023 Climate Change Synthesis Report supports this finding, confirming that tropical coastal regions are receiving disproportionate levels of exposure to extreme weather events based on what is currently known and projected because of climate change [10]. Lastly, Kapukotuwa's independent research into microplastics in Sri Lanka has further validated the complexity and environmental fragility of the Puttalam production environment as a result of requiring that he worked directly with PSS saltern operations [8].

1.1.3. AI and Machine Learning for Agricultural Time-Series Forecasting

Many researchers have demonstrated how useful forecasting-based decision support can be for the agricultural and coastal sectors. Long Short-Term Memory (LSTM) neural networks (a type of recurrent neural network) are effective for predicting agricultural time series in a manner that shares structural similarities with crystallization. Zhang et al. demonstrated that LSTM models do an excellent job of capturing the hydrology of agricultural areas with variable weather by applying sliding window input formats to future meteorological and operational parameters [5]. Additionally, Vrban et al. supported the idea that LSTMs can be used to predict crystallization parameters in seeded-cooling crystallization (a process that shares structural similarities with solar salt pond crystallization) and demonstrated that the model could extract multiple layers of temporal dependencies from the crystallization parameter time series [11].

At first, the most common methods to evaluate forecasts are autoregressive models, such as ARIMA and Holt-Winters exponential smoothing. However, it has been demonstrated by Kumar et al. that these methods do not work well when the time series data are driven by structural/physical processes and not just temporal processes, which has a critical effect on forecast accuracy for solar salt production, for which bed count, rain, and temperature are direct causes of forecasting accuracy [12]. The theoretical framework of Holt's exponential smoothing methods supports this notion, since these methods only decompose previous trend and seasonality patterns from time-varying influences (i.e., stochastic), they will not consider changes in the data-generating process due to changes in structural processes [13]. Another widely used time series decomposition algorithm, Prophet, also shares this shortcoming, as it will decompose historical time series data into trends and seasonality but will not provide a way to incorporate causation.

1.1.4. Statistical Price Forecasting in Commodity Markets

Incorporating external regressors within the classic Seasonal Autoregressive Integrated Moving Average model enables the broader use of this model in both seasonal commodities driven by regular seasonal cycles and in seasonal commodities affected by external (i.e., supply) causes of price change. Banaś and Utnik-Banaś found that models similar to SARIMAX provided good forecasts of short-term prices of agricultural commodities where seasonal exogenous variables strongly affect the structure of prices [6]. Ojha and Karki also found strong evidence that SARIMA-type models could be used to forecast the prices of all types of commodities on the global market, including those affected by climate-related market dynamics [9]. The applicability of this approach to seasonal pricing can be observed from the consistent seasonal demand experienced by premium ("Maha") salt in the PSS wholesale price data, where the seasonal demand by distributors for premium-quality "Maha" and "Yala"-premium seasonal pricing are ascertainable.

1.1.5. Demand Estimation in Agricultural Commodity Settings

Predicting the demand for agricultural commodity products based on historical prices over time is two separate problems from predicting prices in agricultural commodities. Demand in agri-commodity markets will be better represented through use of seasonal fractions of total production than to be independent time series' demand series, a key reason for using the yield ratio methodology presented here. Furthermore, Kumar, et al. provide evidence that autoregressive model estimations of demand in separate production-dependent demand series show very poor time series fit because they do not model a production process.

1.1.6. Computer Vision for Industrial Quality Control

Industrial quality inspection applications for granule and crystalline materials have been benchmarked with powerful performance results using computer vision systems based on deep learning. The YOLO (You Only Look Once) family of object detectors, and specifically, YOLOv8, has become a leading framework for real-time object detection tasks requiring both high throughput and flexible deployment. The YOLOv8 architectural design incorporates a CSPDarknet backbone, PANet feature pyramid neck, as well as decoupled detection heads in such a way as to achieve high-precision detection of multiple object scales on standard computer processing unit (CPU) hardware. Across the adjacent domains of quality inspection of granule products, YOLOv8 has consistently produced mAP@0.5 scores over 0.95 for variants of YOLOv8 and therefore create a compelling benchmark to compare against when evaluating new deployment.

The use of Convolutional Neural Networks (CNNs), such as MobileNet and ResNet, has also been demonstrated in grading food products and agricultural products using an image-level classification approach. This application is limited by its inability to perform spatially localized detection, and thus creates significant limitations in the application of real-time quality grading, which requires per-crystal spatial analysis. The use of the HSV color space for analysis has been validated for providing a highly effective method of determining salt crystal purity during lighting-controlled testing

where the $V \times (1-S)$ whiteness index formula is used as an alternative measure of purity.

1.1.7. Federated Learning for Privacy Preserving Industrial Collaboration

In federated learning, (FL) machine learning models can be trained on distributed data sources, without transferring the data to a central server. This is important for industrial operators who do not want to share their proprietary production data with anyone. The FedAvg aggregation algorithm discussed in many standard/typical forms of the federated learning literature enables common model training across different distributed sources while keeping the data at those sources and giving rise to new forms of data. Differential privacy is also available as part of a federated learning system and provides a formal way for companies to ensure their proprietary business practices are kept private and, therefore, will not leak any valuable production information from their company to another company in their sector.

1.1.8. Microservices Architecture for AI Decision Support Systems

AI decision support systems are increasingly using microservices. Microservices allow independent services (each with their own computational characteristics) to be deployed and/or scaled independently of each other without redeploying the whole system. In a recent systematic literature review done by Ferreira, it was found that the microservice design is most appropriate for AI-composable platform architectures, where some services (such as LSTM) are inference-heavy and need to be scaled independently from other lightweight transactional services [14]. ONNX (Open Neural Network Exchange) Runtime is a production-quality framework that allows trained models from Python-based deep learning products to provide direct inference uses in other programming environments without any communication overhead between different processes — this capability is of relevance in placing a trained BILSTM model into a NestJS/TypeScript microservice designed using the ONNX framework [15].

1.1.9. Gap in the Sri Lankan Solar Salt Literature

A comprehensive literature review of the existing records of AI-enabled forecasting in Agriculture, Fisheries and Coastal Industries indicates that no existing Data driven system exists for predicting crystallizations of solar salt production, and for market intelligence or quality control purposes in the Sri Lankan Solar Salt Industry, nor is there an Integrated Platform or system for forecasting production, planning for market development, assessing visual quality and Distributed Collaborative Learning of production and distribution within a comparable Setting in tropical salt production. As such, the lack of an established Decision Support System in the Solar Salt industry, which contributes directly to National Food Security and provides employment opportunities for thousands of landowners and workers in Puttalam, makes the foundation for developing the BrineX Platform.

1.2. RESEARCH GAP

A systematic overview of existing literature and presently implemented systems reveals that there is no data-driven decision support infrastructure for the solar salt industry in Sri Lanka. There exist numerous gaps in the research that are being addressed by the BrineX Platform which can be grouped into four areas corresponding to BrineX's four modules plus one other broad systemic area that no single gap can cover. These gaps are graphically portrayed in Figure 1.2.1 (with each gap identified being mapped to its corresponding BrineX Module).

- Gap 1 - Absence of Multi-Parameter, Localised Crystallisation Forecasting

Currently there are no systems that integrate multiple daily pond parameters and real-time weather features for the purpose of generating localised, quantitative crystallisation readiness forecasts for Sri Lankan saltworks. Localised forecasts-for example, forecasts produced by agricultural general purpose weather prediction services (e.g. IBM Watson Agriculture)-are not calibrated to the Puttalam microenvironment, do not include daily salinity and brine concentration dynamics, and do not include crystallisation-relevant harvest actions.

At present, saltern operators at Puttalam Salterns rely entirely on their own personal experience and on informal communication with their local peers to make decisions with respect to crystallisation and harvest timing. Given the increasing variability and change in the intensity of the monsoon season described in the 2023–2024 production cycle, relying solely on personal experience and informal communication with local peers has proven insufficient.

Furthermore, none of the existing forecasting systems are able to assist in solving the unique two-stage prediction problem associated with solar salt: 1) predicting crystallisation parameters on a daily basis; and 2) converting crystallisation parameters into production volume forecasts for the following month that have a causal connection to the physical inputs (number of salt beds, amount of rainfall, temperature, etc.). The ARIMA ensemble (the closest available baseline for temporal prediction), using a 36-month series of Puttalam production data, yielded an R squared value of 0.4100; well below the R squared value of 0.90 that would demonstrate that production scheduling decisions could be made based on reliable production forecasts.

- Gap 2 - Absence of Data-Driven Market Intelligence for PSS Landowners

Presently, PSS landowners are faced with three significant decisions regarding production and distribution of their crops. Unfortunately none of these decisions are currently supported by any market knowledge whatsoever. The following two sub-gaps indicate how far away PSS landowners are from acquiring an effective quantity of market knowledge.

There is not an effective method for estimating the expected demand for PSS crops in any given month. Historically, PSS landowners have used informal means or "tip off" sources to estimate demand for their product. Evidence supports that time series models such as SARIMAX when used with disaggregated landowners do not produce satisfactory results in predicting expected demand (i.e., the overall MAPE was 233 percent). In contrast, the quantity of PSS crops produced in any given month appears

to be more closely correlated with the crop being harvested not the timing of when it is harvested (i.e., raw crop production method needs be accounted for in the modeling of expected demand).

PSS wholesale salt prices exhibit a known seasonal price pattern (Maha/Yala), yet landowners do not have an effective method for estimating the price they will receive for their product when they harvest it. Price discovery is controlled solely by intermediaries using an asymmetrical information advantage to deny landowners receiving a fair price for their product. No model has been developed or used prior to this project to forecast prices for wholesale salt in PSS.

There is no formalized process for selecting a purchaser of the timber sold by land owners; purchasers have been selected by land owners as a result of past relationships or transaction history, rather than having been identified through a process of comparing other potential purchasers on the basis of an analysis of how much they would be willing to pay for a parcel of timber. The prior operation of the PSS had no ranked order of purchasers readily available to land owners based on the price per unit and the quantity of timber they could consume or have available for consumption. Therefore, a land owner who successfully harvests timber from their property would need to make all decisions that affect profitability in the market place without any reliable, well-organized information being available.

- Gap 3 - Absence of Automated Quality Control in Tropical Salt Production

In the case of Sri Lankan Salterns, quality assurance is performed strictly through manual inspection done by human beings. The inspectors below ambient lighting conditions visually inspect the crystallized salt to determine if it meets the acceptable quality standards for packaging and subsequent shipment for export or rejection. The result is a slow, subjective, and scalable process that relies upon skilled labour, and the inability of providing the sorts of quantifiable quality documentation required by Export customers and food safety agencies today. To date, no AI provided computer vision system for visual quality control of tropical salt production has been developed or published anywhere globally. However, in related products and food processing industries that sell granular products, deep learned computer vision systems can reliably provide accurate results with precision and recall scores greater than 95%. The data from these systems demonstrates that there is sufficient technical capability to automate inspection for volume, but there are still no systems currently available for salt crystal grading purposes, which have been evaluated using valid existing architectures. An example of such an architecture would be to utilize YOLOv8 for the evaluation of salt crystals. The presence of such a gap in the availability of automated inspection capabilities specifically for evaluating salt crystals constitutes a clear and documentable gap that has both commercial and regulatory implications on Sri Lankan salt export capabilities.

- Gap 4 - Absence of Privacy-Preserving Collaborative Optimisation for Salt Waste Valorisation

Bittern is the byproduct of salt production that contains large volumes of magnesium, potassium and sulfates; however, it is frequently discarded without

recovering any value. Optimising the use of bittern at several different distributed saltern production sites creates additional challenges due to the data-sharing bottleneck created by operators of individual salterns being resistant to sharing their private operational data with anyone else, including a central, industry-wide database and processing system. To date, there is no known federated learning framework that has been published for this industry that supports the collaborative and privacy-preserving optimisation of waste valorisation from salt production. Traditional centralised machine learning approaches cannot be used here because they require transferring data to a central location; therefore, this arrangement is not feasible for technical or business reasons in this industry's context.

- **Overarching Systemic Gap - Absence of an Integrated, End-to-End Platform**

While four-module gaps exist, an even larger one exists at a more fundamental level: no platform has been integrated that combines production forecasts, market intelligence, visual quality control and distributed collaborative learning within any tropical, coastal salt-manufacturing scenario. AI applications are currently being developed independently, PSS must rely on the successful functioning of each of these capabilities as parts of a continuously pleased causal connection. Each step requires the previous step to be properly functioning; for example, crystallization conditions determine the volume of salt produced, which then drives demand forecasts and price expectations that will affect who buys and when the buyer will harvest, which will affect the quality of the salt being harvested (this last connection will affect likely market tier that a given batch of salt will enter). Therefore, there currently exists no published system in the literature that successfully satisfies the entire solar salt manufacturing end-to-end decision support requirement.

Each of the four modules on the BrineX platform—Crystal, Compass, Vision and Valor—have been specifically designed with the intent to fulfil the identified gaps. This is the authors' belief that the BrineX platform will constitute the first published, integrated and data-driven decision support system for the Sri Lankan Solar Salt Industry.

1.3. RESEARCH PROBLEM

This research focuses on developing an integrated, AI-driven ecosystem to provide decision-making support to the PSS Saltern operators and landowners. The ecosystem must support them deciding what is best for their crystallizations forecasting, market planning, quality control, and waste valorization from a causal perspective, structural correctness based on the data-generating processes of the Puttalam solar salt environment, and production validation for implementation in real-world conditions impacted by climate change.

The overarching problem can be looked at as being made up of four smaller Issues that are closely related. There are four separate but related dimensions to the problem. Each of these dimensions adds complexity to the overall assessment of the issue.

These four dimensions consist of:

1. Environmental Uncertainty: The environmental conditions (temperature, rainfall, humidity, wind) that govern the crystallization process, are becoming increasingly variable (due to increases in the variability of South Asian coastal monsoons) making it impossible for experience-based planning to accommodate this level of variability. A data-driven system to forecast the future state of these processes must include relevant pond-level operational performance and use real-time weather data to provide actionable readiness to crystallize signals over a time horizon sufficient for an operational response.
2. Asymmetrical Information in the Market: In this case study, PSS landowners function in an intermediary market where there is no opportunity for PSS landowners to independently discover product demand, price discovery, or identify potential buyers. The research issue related to this dimension is not only an issue concerning the accuracy of predictions of product demand, but also whether there are structural errors in the demand estimation model. A demand estimation model must have a causal relationship established between individual production forecasts to be functionally useful at the individual landowner level. A price forecasting model must reflect the PSS price structure arising from the Maha/Yala seasonal premium in order to provide accurate price forecasts. The buyer recommendation system must provide complete transparency and auditing in order to help restore the information asymmetry at the historical disadvantage to the landowners that existed under the abuse of the intermediary system.
3. QA (Quality Assurance)/RegCo (Regulatory Compliance) - Manual inspection cannot generate the verifiable/reproducible quality records that are now required by export buyers/food safety regulators. The research objective will be to develop a real-time computer vision system that is capable of simultaneously detecting and locating defects/impurities, calculating a composite quality score based on both spectral data and defect/impurity detection results, and interfacing with the overall platform architecture using only commodity hardware without GPU latency associated with traditional inference.
4. Collaboration for Optimization with Data Scarcity - The environmental/economic impact of the disposal of bittern waste is well known; however celebrating all sites across multiple locations regarding valorization optimization has been inhibited because there is no mechanism for sharing data that is compatible with the requirements of industrial secrecy. The research objective in this area is to create a federated learning framework that provides an opportunity for multiple salterns to collaborate on improving models without requiring any saltern operator to share their raw operational data with a central server or any other party.

This comprehensive, multidimensional issue involves addressing both the immediate production crisis revealed in 2023-2024, because of operational vulnerability, as well as the longer-term structural challenges associated with developing a climate resilient, data-driven Sri Lanka salt industry that can maintain national self-sufficiency and export competitiveness in an era of accelerating climate variability.

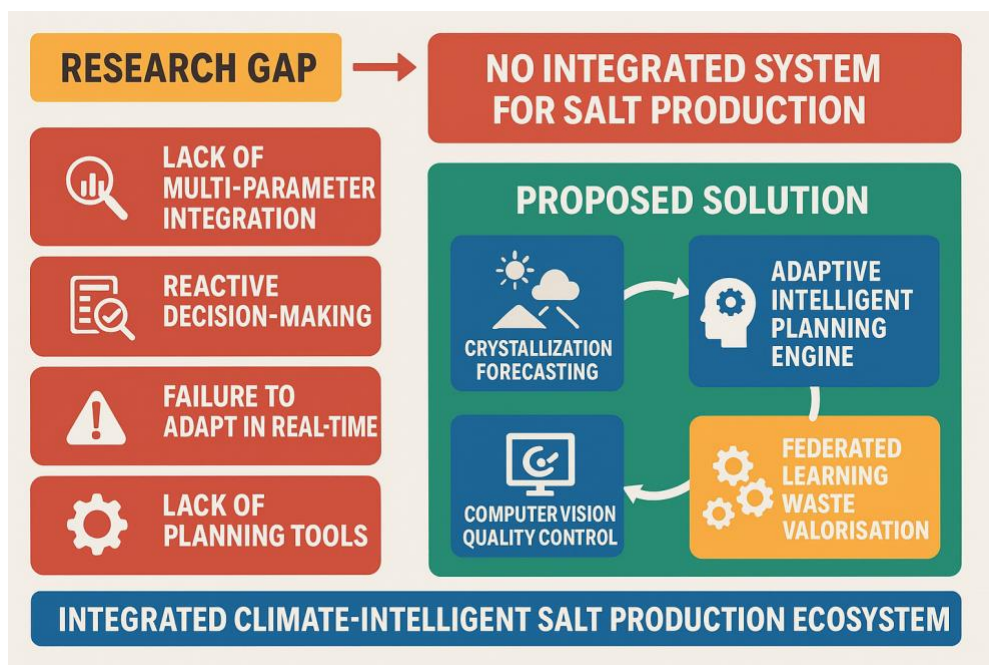


Figure 1.3.1: Diagram to represent Research Gap

2. RESEARCH OBJECTIVES

2.1. MAIN OBJECTIVE

To make BrineX we need to design, develop and put it out there. BrineX is a system that uses intelligence and is on the cloud. We made BrineX with the Puttalam Salt Society. It helps people who make salt by giving them the information they need to make decisions. This includes guessing when salt will form knowing what is happening in the market planning checking the quality of the salt and working together to make the most of waste without sharing information. The goal of BrineX is to help people predict how salt they will make make sure everyone has the same information about the market make sure the salt is good quality and use resources in a way that is good, for the environment even when the weather is changing. We want to do all this with BrineX.

2.2. SPECIFIC OBJECTIVES

The specific objectives below specify the nature and extent of the contributions of each ecosystem component as well as promoting synergy among the components:

- **Crystallization Forecasting:** To develop and validate machine learning models of weather, sensor, and historical production data, and give early and precise deep forecasts of optimal crystallization and harvest windows.
- **Intelligent Planning Engine:** Develop flexible AI-based planning systems that can take predictive forecasts, real-time market, manufacturing, and regulatory information, and convert it into stakeholder-optimized, actionable operational plans to make profitable, proactive decisions.
- **Computer Vision Quality Control:** Implement a high-throughput automated computer vision system to grade the salt and detect salt impurities in real-time so that the high-quality and regulatory standards of the product can be achieved, and any human touch and production bottlenecks are reduced.
- **Federated Learning Waste Valorization:** Develop a federated learning infrastructure, a secure and collaborative setting to optimize the reuse and valorization of salt waste in a distributed set of sites, without compromising data that is sensitive but also enhancing a sustainable circular and economic enrichment initiative.

In attaining these goals, the project will not only assist Sri Lanka in the self-sufficiency in salt production, competitiveness in both regional and international markets, and sustainable and technology-driven management of its resources but also helping provide the country with the necessary leadership in the above aspects. The replicability of the ecosystem in terms of modularity and scalability also allows the ecosystem to be implemented in other tropical coastal nations that are grappling with the same challenges and thereby making more contributions to the topic of climate-smart industry transformation.

3. METHODOLOGY

3.1. System Overview

The Integrated Climate-Intelligent Salt Production Ecosystem is a modular, AI-driven platform (potentially modular) that aims to revolutionize the salt production industry in Sri Lanka by performing end-to-end decision support tasks to the entire salt production lifecycle. The structure of its architecture is divided into four related components that take care of each of the vital steps in operational planning, execution, and optimization.

System Components

1. **Crystallization Forecasting:** Combines multimodal information (weather, other) with advanced time series models (ARIMA, Prophet, and LSTM) to determine the ideal conditions to crystallize salt and harvest it. This makes it possible to develop proactive plans and to reduce the risks of yields depending on the uncertainty of the climate.
2. **Intelligent Planning Algorithm:** Converts predictive intelligence of Component 1, with real-time market trends, production capacities, economic data, and regulatory requirements into optimized operating strategies implemented into actions. It uses multi-parameter optimization, heuristics, fuzzy logic, decision trees, and reinforcement learning to support dynamic data-driven planning, which can change as circumstances change.
3. **Quality Control: Computer Vision:** Automatically grades the salt quality using computer vision image analysis and impurities are detected in real time. This improves management without the need of manual check-up, thereby making it more regulated and assuring the quality of the products to both the local and international markets.
4. **Federated Learning Waste Valorization:** Facilitates the distributed and privacy-preserving coordination of different sites of the saltern to optimize by-product reuse and valorization. By using federated learning and differential privacy, they can be sure that critical operational data will not be shared directly, but it will lead to innovation and data safety.

Key Features

- Modularity
- Real-time Data
- User-Centric Dashboard

Such an interconnected ecosystem has the unique opportunity to consider the complex climate sensitive operation realities of the salt industry in Sri Lanka.

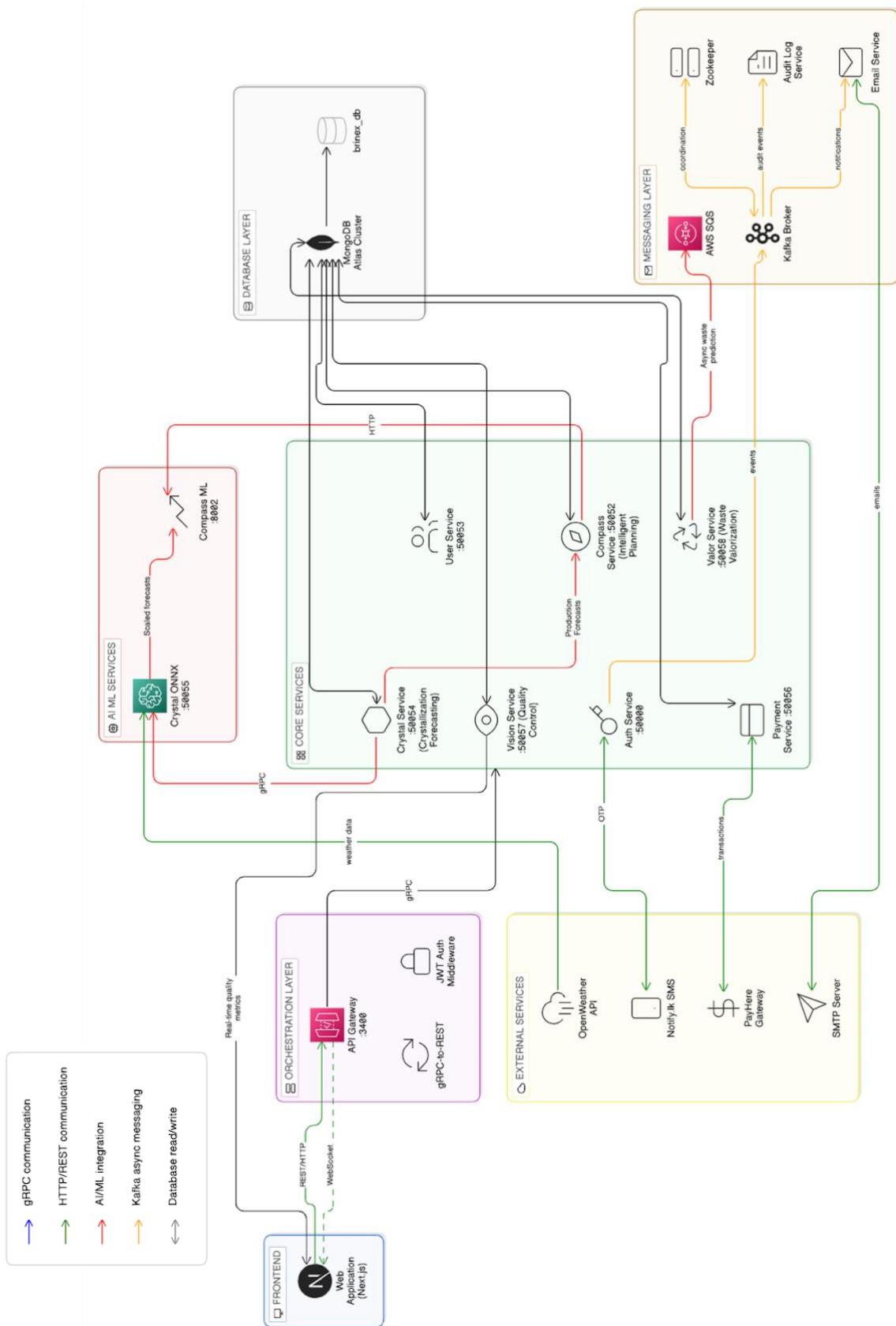


Figure 3.1.1: System Diagram

3.2. Technology Stack

This platform is constructed using a combined technology stack of NestJS/TypeScript and Python/FastAPI. The transactional operations of NestJS microservices facilitate CRUD capabilities for daily pond measurement; buyer offer management for the lifecycle of a harvest plan; and ONNX Runtime inference pipeline for Crystal. The ML operations of Python/FastAPI microservices are utilized for the statistically intensive operations in running the SARIMAX price forecasting model; the yield ratio demand algorithm; and the seller ranking engine in Compass. All services are containerized using Docker and managed through Docker Compose for local and staging deployment, with AWS being the target cloud infrastructure.

Inter-service communication is accomplished using gRPC, which employs Protocol Buffer contracts as the basis for all synchronous service-to-service calls. This process guarantees strongly typed communication between modules via low latency (non-blocking) and high throughput channels. Apache Kafka is used for asynchronous event processing for non-blocking operations such as quality inspection result broadcasting and coordination of federated model updates. MongoDB Atlas serves as the unified persistence layer for all services. Ten (10) CI/CD pipelines in GitHub Actions enable independent builds and testing of each service through path-filtered triggers to ensure that changes to one module do not cause unnecessary rebuilds of any other modules.

3.3. Data Flow Architecture

The BrineX platform's end-to-end data flow starts with saltern operators at PSS entering daily pond measurement readings (eight crystallisation parameters) into the Crystal service, which also collects fourteen real-time weather factors (from OpenWeatherMap API). PSS produces crystallisation parameter forecasts for the next 60 days with the help of an input data dual-input Bidirectional Long-Short Term Memory (LSTM) model that runs within the NestJS Microservice using a prediction engine called ONNX Runtime.

These daily forecasts are provided as inputs to a linear regression model that converts the daily forecast to monthly productions scaled to the participant's number of beds. These production forecasts are then sent to Compass using gRPC and an HTTP bridge, and the Crystal Nest services acts as a gRPC client to Compass' FastAPI based ML service via HTTP. Compass receives the production volumes in bags and scales them to the participant in three different ways: demand algorithm, SARIMAX price forecasting, and ranked purchaser system. The combined outputs from these methods will appear on the BrineX harvest plan dashboard allowing for data-driven decision-making regarding harvest timing, pricing strategy and purchaser selection.

Vision is also processing the live video feed from the saltern inspection station through the YOLOv8m model, which is running using ONNX Runtime as a microservice in NestJS. The YOLOv8m model detects and classifies (salt crystal) in real-time and produces individual crystal whiteness indices and collective quality scores for each detected crystal. The results of this detection are sent to the operator dashboard over WebSocket, allowing the operator to continuously monitor the quality of product while also accessing the daily production and market planning output from both Crystal and Compass. The Valor platform operates as a separate distributed

coordination layer for federated learning models and all saltern locations update their individual models locally and independently on their operational data. FedSalt aggregates these model updates for the participating saltern locations using the FedAvg protocol and sends the model updates to an Amazon SQS asynchronous message queue without requiring any site to share their individual operational data. The FedAvg model update is then sent back to each participating saltern location to facilitate collaborative optimising of bittern valorisation throughout the entire PSS network.

3.4. Module 01: Crystal - Crystallization Forecasting

3.4.1. Requirement Gathering and System Design

The first step of the dual-phase production-to-market intelligence pipeline of BrineX (Crystal) is the production forecasting tool (Crystal) that was built based on input from PSS saltern operators. The operational environment of a saltern is the primary influencer when it comes to determining the requirements for this system. These system requirements include: a forecast horizon of 60 days, which is an adequate lead time for harvest planning; daily granularity forecasted due to the day-to-day variability of crystallisation conditions; and forecast confidence bounds as ways to represent uncertainty in the forecast vs using any form of a false, single value forecast; an accessible dashboard and interface to the daily operation crew members (non-data science background) so that they can review forecast information.

Crystal has been developed as an open-source NestJS/TypeScript microservice that provides RESTful API endpoints for the inclusion of different types of daily data related to saltern operational parameters, such as daily measurements taken, daily crystallisation parameters, monthly production predictions and performance statistics generated by the model used to generate these outputs. By embedding the LSTM forecasting model using ONNX Runtime within the microservice for execution requests, crystal provides an efficient way of making predictions in native TypeScript format without having to cross languages and deal with process overhead (subprocess events) to and from different languages. To provide production forecasts to the Compass service downstream, the Crystal service sends production forecasts using a gRPC contract as a gRPC client to the Compass service (FastAPI) over HTTP.

3.4.2. Dataset

The crystallisation dataset consists of 1,095 daily observations from January 2023 to December 2025, collected directly from PSS saltworks' crystallisation ponds. Each observation involves eight measurements about the pond, such as brine salinity, pond depth, temperature, and indicators of crystallisation rates, along with 14 weather measurements from the OpenWeatherMap API: for example, rain, humidity, wind speed, and temperature. All 994 missing weather measurements were filled using a forward-fill method. Pond parameter and weather parameter data were scaled independently using separate RobustScalers so that neither was affected by the other's scale.

The production and pricing dataset contain 36 months of monthly production volumes and wholesale salt prices from PSS, the same source that provided the crystallisation dataset; period of time is also from January 2023 to December 2025.

3.4.3. Data Processing and Feature Engineering

The daily logs were processed into a series of 60-day overlapping sliding windows to provide input tensors to the Bidirectional LSTM. The initial training data set was expanded by generating sequences at three-day intervals. This resulted in creating 3,285 training samples over the entire 1,095-day record, three times the number of original records generated. This augmentation of the records was essential given that there were only three years of operation at the time of development. Therefore, the data sets were split into training (2,217), validation (474), and test (475) samples without changing their respective order or introducing any data leakage.

For the Linear Regression production calibration, monthly aggregates from the 36 months of PSS production data were used. In addition to the 36 months of data collected for the PSS production dataset, four causal features were created to be used as explicit regressors: salt bed count (the primary structural driver of production capacity), total monthly rainfall (primary climate-driven disruption), average monthly temperature (positive driver of evaporative rates and crystallizations efficiencies), and seasonal encoding using sine/cosine transformations of the month of the calendar year in order to account for the Maha/Yala seasonal cycle and to avoid imposing a linear time trend.

3.4.4. Forecasting Model Development: Multi-Model Approach

To complete model selection, Crystal went through a rigorous process of evaluating five possible models before deciding upon one to deploy as its final solution. One initial choice was ARIMA (Autoregressive Integrated Moving Average) being used as a baseline comparison; it is widely used in the industry, hence it was deemed the appropriate time-series production forecasting method against which to measure other methods.

Next, Holt-Winters exponential smoothing and Prophet were both given consideration as alternative methods of statistical forecasting techniques based upon regression analyses. Additionally, Random Forest and XGBoost were chosen as candidate methods of building non-sequential ensembles producing results from all of the statistical prototype models mentioned herein.

Ultimately, all five potential model alternatives produced either sub-optimal predictions utilizing the 36-month PSS (production, shipment and sales) historical dataset; and/or fundamentally incompatible structures for producing forecasts based upon the causal relationship inherent in solar salt production.

The inherent limitation of ARIMA was validated: when applied to the same 36-month PSS production data, the ARIMA ensemble (with 10 models) yielded $R^2=0.4100$, far short of the $R^2=0.90$ operational reliability standard for planning purposes. More importantly, the ARIMA forecast standard deviation of 2,254 bags (compared to a standard deviation of historical production of 43,605 bags) indicated that the model was reducing the forecast to a constant (essentially, it was not

accounting for the seasonal variability in PSS production). Holt-Winters and Prophet suffered from the same structural constraint - neither model has a way to include bed count additions or rainfall anomalies as causal factors, meaning that any change in the production environment is not learned by the model until new production data is gathered over the next few months.

3.4.5. Selected Architecture: Dual-Input Bidirectional LSTM

A two-input Bidirectional LSTM network is chosen to take pond measurements and weather features as inputs to two branches, merge their encoded representations, and create 60-day forecasts of all eight crystallisation parameters in a single inference.

The pond measurement branch takes a (60, 8) input sequence and passes it through two Bidirectional LSTM layers with 256 and 128 units, respectively. The weather branch takes a (60, 14) input sequence and passes it through two Bidirectional LSTM layers of 64 and 32 units. Each Bidirectional layer is followed by Batch Normalisation and Dropout (rate=0.3) layers to normalise and regularise the training process. The branches are combined to produce a 192-dimensional representation which is then processed through three Dense layers with 256, 128, and 480 units. The output layer produces 480 values, reshaped to (60, 8) which corresponds to 60-day-ahead predictions of all 8 crystallisation parameters per inference run.

Because the retrospective and concurrent parameter forecasting architecture is capable of representing two types of trends at the same time-forward temporal trends, e.g., salinity increasing during dry concentration and backward contextual dependencies, e.g., the residual impact of previous rainfall on present day salinity level of ponds-it's an ideal architecture for forecasting crystalise parameters. This dual input graph architecture distinctly separates the operation (pond) and environmental (weather) signal/output streams; thus, the neural network is able to learn the unique temporal dynamics associated with each signal type independently prior to being integrated into a unified representation.

The final architecture has 532,832 parameters and was compiled using the Adam optimizer with a learning rate of 0.001 and mean square error (MSE) as the loss function. The training was performed over 150 epochs, with early stopping occurring at approximately the 120th epoch. Once trained, the model was serialised to be in ONNX format and deployed using ONNX Runtime in the NestJS micro-service, which allowed for inference as a native TypeScript without the overhead of cross-language subprocesses.

3.4.6. Production Calibration: Linear Regression

A causal linear regression model was built to convert daily forecasts of the crystallisation parameters from LSTM into monthly production volume forecasts. The regression model was calibrated using 36 months of production data from PSS facilities and identifies four causal independent variables as exogenous regressors. The formula for estimating production can be represented as follows:

$$\text{Production} = 35.2153 \times \text{beds} - 33.9257 \times \text{rain} + 3763.3592 \times \text{temp} - 279846.18$$

In the above equation, “beds” refers to the number of salt beds, “rain” refers to the sum of all rainfall in a month (in mm), and “temp” refers to the mean temperature in a month (in °C).

This causal relationship directly identifies the principal weakness of using autoregressive models in this instance. The production forecast will be updated when a landowner builds new salt beds with no need for additional historical data. When monsoon /abnormal rain does not conform to historical data, the rainfall will communicate its significance. The model accounts for 97.32% of the variability in monthly production volumes ($R^2 = .9732$) across the entire 36 month PSS database compared to $R^2 = .4100$ of the ARIMA ensemble on the identical data.

A key design consideration relates to the scale of the Linear Regression (LR) model: it was fit to PSS facility-level data for 7,500-10,000 beds. Landowners in the PSS typically have 20-45 beds. Using the facility-level coefficients to estimate production for individual landowners generates invalid results. A two-stage scaling approach was, therefore, adopted: the facility-level model forecasts production, and this is scaled to individual landowners in proportion to their bed count as a ratio of total PSS bed capacity, thereby maintaining the causal structure of the model while generating operationally relevant per-landowner forecasts.

3.4.7. Model Training, Tuning, and Evaluation

The training of the Bidirectional LSTM was performed using 3,285 augmented samples that were divided into 2,217 for training, 474 for validation, and 475 for the test. The Adam optimizer was selected to use with a 0.001 learning rate based on results obtained from the grid search which included learning rates of 0.0001, 0.001, and 0.01. As well, a dropout rate of 0.3 was selected as the regularisation parameter based on testing dropout rates of 0.2, 0.3, and 0.4. MAPE was not included in LSTM evaluation metrics because during inactive period(s) of the pond, certain crystallisation parameters had values near zero which resulted in the mathematical breakdown of the percentage-error calculation for percentage errors. Final evaluation metrics used were R^2 , RMSE, and MAE computed on RobustScaler-normalised values.

The Linear Regression model was evaluated by the in-sample R^2 and MAE for the entire data set of 36 months and by the holdout R^2 and MAE on the most recent six months held out of training. Backtest coverage was established for all 35 normal months of operation by calculating backtest coverage against the 95% prediction interval of $\pm 15,739$ bags.

3.4.8. Dashboard and Visualisation

The Crystal operator dashboard, developed in Next.js, provides four key views: a daily crystallisation readiness panel with the current-day forecast for all eight crystallisation parameters with prediction intervals; a traffic-light alert system (Red/Yellow/Green) that highlights days where forecast conditions are not suitable for crystallisation; a seasonal production outlook panel that shows monthly production forecasts for the next two months with prediction intervals; and a historical performance panel that displays backtest accuracy and other model health indicators. The dashboard is

intended for use by saltern operators who lack data science skills, and uses natural language and colour to indicate alerts rather than statistical metrics.

3.4.9. Testing and Validation

The Crystal service was unit and integration tested with 4 test suites, 65 tests, with 100% (0 failures) passing. End-to-end testing demonstrated correctness of 17 endpoints supporting daily measurement CRUD, daily crystallisation parameter prediction, monthly production prediction and model performance. Load testing was done using Grafana k6 with three test scenarios:

Table 3.4.9.1 : Crystal - Performance Test Results

Scenario	Total Requests	Failed Requests	Avg Response (ms)	P95 (ms)	Check Pass Rate
Smoke (1 VU)	48	0	183.90	488.17	100%
Average Load (5 VUs)	728	0	92.36	153.74	100%
Stress (15 VUs)	3,112	0	88.80	146.87	100%

The unexpected reduction in average response time from 183.90ms for smoke to 88.80ms for stress is a result of proper connection pooling and optimised ONNX Runtime inference under stress. The absence of failed requests in all three scenarios (including 3,112 stress-test calls) validates readiness for deployment.

3.5. Module 2: Compass - Market Intelligence and Planning

3.5.1. Requirement Gathering and System Design

The Compass module of BrineX acts as a "guide" for production-to-market analytics; it does not provide any services directly to the PSS landowner; instead it provides them with historical and future analytic information through which the landowner can make informed decisions about what to produce, how to price it, and the best partner(s) based on their offered terms. Compass is developed as a microservice using the Python/FastAPI framework that is integrated into the entire BrineX ecosystem of 13 microservices. All data related to production volume is transferred from Crystal to Compass using the gRPC to HTTP bridge methodology, where the Crystal NestJS service acts as the gRPC client to the Compass FastAPI service using HTTP. Production volume (in bags, scaled to each landowner's bed count) serves as the main input into all three Compass functions or analysis types. Buyer offer data is stored in MongoDB Atlas. Both the Compass user interface and the Compass api are secured via role-based access permissions. The functional requirements for the Compass design can be found in Table 3.5.1.1.

Table 3.5.1.2: Compass - Functional Requirements

ID	Requirement
FR-C1	Accept monthly production forecasts from Crystal via gRPC-to-HTTP bridge
FR-C2	Compute per-landowner demand estimates using seasonal yield ratios
FR-C3	Generate two-month price forecasts (month+1, month+2) via SARIMAX
FR-C4	Rank registered buyers by price_per_bag descending, filtered by minimum bag_count
FR-C5	Surface demand estimate, price forecast, and ranked seller list via harvest plan dashboard
FR-C6	Store buyer offer records in MongoDB Atlas
FR-C7	Support harvest plan lifecycle (create, query, error handling) via REST endpoints
FR-C8	Provide combined demand-price query endpoint for integrated plan generation
FR-C9	Automatically upgrade to farm-specific yield ratios as transaction history accumulates
FR-C10	Return SARIMAX model performance metrics on request

The non-functional requirements governing Compass performance, availability, and auditability are listed in Table 3.5.1.3.

Table 3.5.1.4: Compass - Non Functional Requirements

Category	Requirement
Performance	Average response time under average load (5 VUs) must not exceed 1,200ms
Availability	99% uptime to support daily harvest planning decisions
Scalability	Handle concurrent planning requests from multiple landowners without degradation
Security	All inter-service communication uses Protocol Buffer contracts over gRPC; REST endpoints secured via role-based access control
Maintainability	Modular FastAPI architecture enabling independent redeployment without disrupting other BrineX services
Auditability	Seller ranking must be fully explainable — each buyer's position traceable to price_per_bag and bag_count
Interoperability	Must consume Crystal forecasts via the standardised gRPC-to-HTTP bridge without Crystal-side modifications

3.5.2. Demand Forecasting - Yield Ratio Algorithm

The demand estimation method known as the Compass is built on a fundamental structural characteristic of the PSS saltern market: demand is production-driven and not time-driven. The owner of a PSS saltern produces salt throughout the year but sells only part of what they produce at any given time during a given season.

Distributors prefer to buy during the Maha season when the greatest availability of dry crystallisation conditions creates the best-quality salt. Therefore, demand is ultimately determined by production, and it follows that price is a downstream effect of the production conditions rather than being determined autonomously as a time-series forecasting event.

Because of this structural characteristic, any time-series forecasting methods – including SARIMA with exogenous variables (SARIMAX) – must be rejected as an adequate method to estimate demand. For example, when SARIMAX was used to forecast disaggregate demand per landowner based on historical data, it resulted in a mean absolute percentage error (MAPE) of 233% confirming that when demand is modelled as a time-dependent series at an individual landowner basis that is inconsistent with the structural nature of the PSS market. Conversely, using the yield ratio method to forecast demand provides an estimated MAPE value of approximately 10.0% based on a hold-out set using actual PSS landowner transaction data, thus providing a 20 times smaller average forecast error than using the SARIMAX model for each landowner.

The yield ratios for each season were computed using 36 months of actual PSS operational data to provide the average demand to production ratio for each seasonal period.

Table 3.5.2.1: Compass - Seasonal Yield Ratios

Season	Yield Ratio	Basis
Maha (peak crystallisation, Dec–Mar)	10.27% (0.1027)	36-month PSS mean
Yala (secondary season, May–Sep)	8.74% (0.0874)	36-month PSS mean
Transition (inter-seasonal, Oct–Nov)	9.51% (0.0951)	36-month PSS mean

Per-landowner demand for month M is calculated as:

$$\text{Demand}(M) = \text{Production_Forecast}(M) \times \text{Yield_Ratio}(\text{Season_of_M})$$

The formula establishes a causal flow: pond conditions → LSTM crystallisation prediction → Linear Regression production forecast → Compass demand estimate. The whole sequence is transparent, explainable and actionable at the landowner level.

The system also has an automatic upgrade: as data on past transactions is gathered after system deployment, farm-specific yield ratios will be automatically substituted for the regional yield constants, thus improving the demand estimate over time without having to change the code.

Rejected Alternatives - SARIMAX for Demand:

SARIMAX on aggregate PSS demand data had a 8.6% MAPE on held-out data, better than the yield ratio algorithm's 10.0% MAPE. But SARIMAX was rejected on theoretical grounds. SARIMAX is an autonomous time series that does not know individual production volumes. It cannot tell the difference between a landowner producing 20,000 bags in a given month and a landowner producing 80,000 bags in the same month. This renders it structurally incapable of producing demand estimates at the landowner level, even if it's correct at the aggregate level. The principle used

here (and throughout BrineX model selection) is to favour structurally correct models over marginally better-performing aggregate metric models when the lower-error model is not capable of describing the process that generated the data.

3.5.3. Price Forecasting - SARIMAX Model

The Compass salt price forecast is based on a seasonal autoregressive moving average (SARIMAX) model (1,1,1)(0,1,1,12) applied to 36 months of monthly PSS wholesale price data, using the Python statsmodels library. It has four additional regressors identified based on structural considerations of PSS price dynamics:

- `is_maha_season` - binary variable to capture the price premium paid by distributors for high-quality Maha-season salt which is crystallised under drier conditions
- `price_lag_12m` - lagged price by 12 months to account for the high annual seasonality in the autocorrelation of price
- `price_lag_1m` - lagged price by 1 month to capture the month-on-month persistence in price
- `total_production_bags` - total PSS bags produced to capture the volume-price sensitivity of local salt market

The implicit differencing ($d=1$, $D=1$) accounts for price non-stationarity. The absence of residual autocorrelation (Ljung-Box test at lag 12) after fitting the model suggests the model is well-specified.

The two-pass recursive method used to generate month+2 forecasts is described below. Pass one creates the forecast for month+1, then, in pass two, the forecast produced in month+1 replaces `price_lag_1m` to compute the month+2 forecast. This two-pass procedure eliminates the potential for data leakage, as `price_lag_1m` at month+1 only uses the forecast from the previous month. The model was validated using rolling-origin cross-validation over the entire sample of 36 months.

Using directional accuracy, the model's directional accuracy currently stands at 46.3%, which falls short of the random baseline of 50%. This result is operationally expected and does not indicate model failure. PSS salt prices tend to exhibit mean-reverting behaviours, fluctuating about seasonal averages but not trending persistently one way. Accordingly, PSS landowners do not need directional signals to make decisions about harvesting and contracting; they mainly rely on price estimates at specific points in time to assess if harvesting under current market conditions makes sense and to establish a defensible position for opening negotiations with distributors over pricing. Furthermore, since the mean absolute percentage error (MAPE) at month+1 of 1.16% directly provides the PSS landowners with price estimates, it effectively ends the informal price discovery process that has occurred as a function of intermediaries using the information advantage to exploit the PSS landowners.

3.5.4. Seller Recommendation - Rule-Based Ranking

The recommendation system for sellers selects the top ranked buyers with submitted bag counts (versions of `bag_count`) that equal or exceed the `price_per_bag`

minimum of the seller as defined in the harvest plan. It accomplishes this via the following steps:

1. From all submitted bag_counts, select only those that equal or exceed the minimum price_per_bag (the threshold).
2. Sort the minimum price_per_bag bag counts in descending order of price_per_bag.
3. Display the sorted list in the BrineX harvest plan dashboard.

The seller recommendation system is 100% transparent and auditable because each buyer's position on the seller recommendation list can be traced back to their submitted bag_count and price_per_bag information. The system does not use any secret parameters, weights or criteria. The seller recommendation system design went through two rounds of assessment and elimination of machine learning based options prior to getting to this design.

Table 3.5.4.1: Compass - Seller Recommendation Approach Comparison

Approach	Top-1 Accuracy	Decision	Reason
Reinforcement Learning	13.5%	Rejected	Structural mismatch — RL is a sequential, stateful framework; seller recommendation is a stateless, single-episode problem
CatBoost	67.5% (97.3% Top-3)	Rejected	Unnecessary complexity — the correct answer (highest price per bag meeting volume requirements) does not require a learned model; introducing one obfuscates ranking criteria
Rule-Based Ranking (deployed)	Optimal by construction	Deployed	Fully transparent, fully auditable, directly addresses information asymmetry

The CatBoost result is illustrative: with a 97.3% Top-3 accuracy, it empirically verified the recommendation task has a transparent solution that is not a machine learning problem. We did not reject it because the ML approach didn't work, but because the most obvious solution is the right one. In a market where PSS landowners have been traditionally short-changed by intermediary opacity, it would not be consistent with the aims of the module to swap one form of opacity (intermediary manipulation) for another (black-box model).

3.5.5. Testing and Validation

The Compass planning service has been unit and integration tested over 3 test suites with 15 tests, of which zero failed. Integration testing confirmed 13 endpoints for the harvest plan life cycle, harvest plan query, errors, and demand/price query. Load testing was performed with Grafana k6 in three scenarios:

Table 3.5.5.1 : Compass - Performance Test Results

Scenario	Total Requests	Avg Response (ms)	Check Pass Rate
Smoke (1 VU)	40	212.78	97.5%
Average Load (5 VUs)	318	1,160.61	96.2%
Stress (15 VUs)	710	2,376.35	86.9%

The total number of failed requests (28 for smoke, 225 for average load, 505 for stress) is reflective of the k6 test script, which tests for both happy-path and scripted 4xx error-handling scenarios. These are not service failures. The stress-load performance loss in average response time (2,376ms) is due to unoptimised database queries on the Deals and Offers endpoints in MongoDB Atlas. The engineering fix is to implement index optimisation on these endpoints; this does not imply any issues with the Compass algorithms.

3.6. Module 3: Vision – Real Time Quality Control

3.6.1. Requirement Gathering and System Design

The BrineX Vision module provides automated detection of salt impurities and multi-parameter quality evaluation that is currently done manually by humans at PSS salterns. Vision can operate on a standard CPU hardware platform in real time and sends per-crystal quality evaluations and batch statistics (to an operator dashboard) without requiring GPU-based systems.

Vision is implemented as a NestJS micro-service that uses a gRPC API to wrap the YOLOv8m ONNX model and MongoDB Atlas as a persistence layer. The real-time inspection results are sent to the operator's dashboard using a WebSocket channel. There are three operating categories identified: pure salt, impure salt, and unwanted foreign materials.

3.6.2. Dataset Collection and Annotation

The Vision training dataset consists of 1,952 images of salt crystals taken in the environment of Puttalam salterns. A camera and LED lighting were used to ensure consistent lighting conditions. Images were labelled using the Computer Vision Annotation Tool (CVAT), and each image was annotated for the bounding box and class label, corresponding to the three operational classes.

Data augmentation was applied to enhance generalisation and the final training dataset is outlined in Table 3.6.2.1.

Table 3.6.2.1: Vision - Dataset Configuration

Split	Images
Training	Augmented from 1,952 base images
Validation	Held-out partition
Test	293 images

3.6.3. Model Selection

The vision model was selected in a pilot project that evaluated three models: MobileNet, ResNet-based image-level classifier, and YOLOv8. MobileNet and ResNet were trialled as image classifiers. Neither could be used because image classification does not provide spatial localisation of crystal impurities within the image - a requirement for real-time inspection, where multiple crystals of varying quality may be present within a single image. We chose to deploy YOLOv8 because it offers simultaneous detection and localisation of multiple object scales within a single forward pass, consistent with the real-time throughput constraints on CPU hardware.

The medium-sized YOLOv8 (YOLOv8m) model was chosen over the smaller YOLOv8n and YOLOv8s models on the basis of the accuracy-latency trade-off: the medium model achieved significantly better MAPE 0.5:0.95 at a throughput of five frames per second on commodity x86 hardware, which was adequate to satisfy the real-time inspection requirement without needing GPU resources.

3.6.4. YOLOv8m Architecture

YOLOv8m uses a CSPDarknet backbone for multi-scale feature extraction, Path Aggregation Network (PANet) neck for generating a feature pyramid across different detection scales, and decoupled detection heads for classification and bounding box regression. Spatial Pyramid Pooling - Fast (SPPF) is used in the backbone to pool multi-scale spatial information. Post-detection Non-Maximum Suppression (NMS) is used to remove redundant detections. The network was trained end-to-end for 150 epochs on the dataset with early stopping applied at around 120 epochs as plateau in validation MAPE 0.5 became noticeable.

3.6.5. Quality Metrics Formulation

Vision produces three metrics for each analysed frame:

- Purity Percentage - The number of detected boxes in the frame tagged as pure salt over the number of all detections in the frame, directly giving the batch purity percentage.
- HSV Whiteness Index - measured for each detected crystal as $V \times (1 - S)$ with V being the Value (brightness) channel and S the Saturation channel of the crystal's hue, saturation, value (HSV) colour representation. This results in a score close to 1 for pure white crystals (bright and unsaturated) and close to 0 for coloured and impure crystals.
- Composite Quality Score - a class weighted score that combines the spectral purity (HSV whiteness index) and detection outcome (class label) into a single score. The composite score offers a single metric that aggregates the visual appearances and detection-based classification of each batch, and can be used as the quality gate to determine whether a batch is accepted or rejected for harvest.

3.6.6. ONNX Export and Deployment

Following training, the YOLOv8m model was exported to ONNX format for runtime-independent deployment. The ONNX model is served via ONNX Runtime within the NestJS microservice, enabling native TypeScript inference. Input tensors are formatted in NCHW layout (batch, channels, height, width). Region of Interest (ROI) filtering is applied pre-inference to restrict detection to the active inspection area, reducing false positives from frame edges and background elements.

3.6.7. Dashboard and Operator Interface

The Vision operator dashboard, built in Next.js with WebSocket streaming, provides four primary views: a real-time quality inspection screen showing live camera feed with bounding box overlay colour-coded by class (pure in cyan, impure in dark blue, unwanted in white), a purity gauge, and batch statistics; an alert history screen showing filtered rejection events with timestamps, severity levels, and resolution status; a batch purity trend chart showing the 24-hour area chart of purity score evolution; and a system health panel showing inference latency and throughput metrics.

3.7. Module 4: Valor - Federated Learning for Waste Valorization

3.7.1. Requirement Gathering and System Design

The Valor system addresses the issue of salt waste valorisation through the deployment of distributed PSS (Partial Salt Solutions) saltern sites. In the process of producing solar salt, large quantities of end-products created in the process known as 'Bittern' (a mineral-rich by-product containing substantial amounts of magnesium, potassium, and sulphites) are produced but disposed of as a waste product without any mineral recovery.

To optimise bittern's future use, collaborative network modelling is essential across multiple sites; however, historical data-sharing barriers between companies (primarily from individual companies' reluctance to disclose sensitive or confidential production data) have precluded the utilisation of traditional, centralised methods of modelling this collaboratively.

In order to overcome the data-sharing barriers, Valor has developed a new approach to modelling using a privacy-preserving federated learning method that enables each of the saltern sites involved in the project to independently train a local model based on their own site's actual production data while allowing a single central aggregation server to combine all of the trained local models into one final overall model using the FedAvg algorithm without ever accessing each site's raw production data.

In addition, a combination of Amazon SQS (Simple Queue Services) as an asynchronous message queue between each saltern site and the central aggregation and communication server, makes the entire solution very resilient to the challenges associated with intermittent network connectivity issues that typically present themselves in remote and offshore coastal saltern sites.

3.7.2. Dataset and Digital Twin Approach

There are not enough examples of how these particular types of saltworks operate to train a reliable ML model about that site by itself. Valor's solution to the "cold start" data limitation is to apply a thermodynamically-informed digital twin concept: we created a 27-year synthetic dataset of evaporation efficiency indices from physics-based calculations plus random variability using stochastic uncertainty modelling to create an operating dataset that includes the complete range of operational and environmental variability that could be experienced at Puttalam, as well as observe physical limits on evaporation and crystallisation.

3.7.3. Model Architecture

To develop the FedSalt Model, we leveraged an ensemble of Deep Neural Networks and Gradient-Boosted Learners (model v2) to predict elemental ion concentrations in bittern (e.g., magnesium [Mg], potassium [K], sulphate [SO₄]) and thus provide the data necessary to support valorisation planning. The combined power of deep NN capabilities to model complex non-linear interactions in features and the inherent strength of GBT methods for modelling tabular operational data results in a stable, more accurate predicted value for both total Mg, K and SO₄ concentrations when compared against the predicted value from either model type separately.

3.7.4. Federated Training Protocol

Local model updates are generated from localized operational data, at each saltern site, through a training process called Federated Training. These updates consist of gradients (and/or model weight deltas) as opposed to actual data and are communicated to the FedSalt Aggregation Server via Amazon SQS. The FedSalt Aggregation Server utilizes the FedAvg algorithm to compute the overall model based on all local updates, which is subsequently sent out to the individual sites for the next training cycle. The use of Amazon SQS ensures that the global aggregation round does not block any sites with intermittent connectivity, which is vital to successful remote coastal saltern operations.

3.8. Commercialization

3.8.1. Value Proposition

BrineX delivers value on three fronts. First, at an individual landowner level, Compass provides price-level forecasts and transparency in buyer ranking that corrects the information imbalance that intermediaries used to exploit while providing a direct advantage to the landowner for negotiating. Secondly, PSS's Crystal provides the production forecasting and Vision provides the quality-control infrastructure necessary to facilitate participation in formal export markets at the institutional level. Finally, at a national scale, the increased production predictability, the quality assurances supplied through the reporting and the development of a better-informed, market-oriented Salz industry will decrease Sri Lanka's reliance on salt imports and contribute to strengthening the country's food security.

3.8.2. Revenue Model - SaaS Tiered

The BrineX platform offers three tiers of subscription; Basic, Professional, and Enterprise.

- The Basic Tier allows for customers to enter their manual measurements into the platform, receive monthly forecasts of demand and pricing trends, and is directed towards individuals who own or operate less than 50 beds. The monthly fee is low enough that it will provide small subsistence scale operators access to Crystal and Compass production and pricing information.
- The Professional Tier provides real time pricing forecasts, provides sellers with a full ranking of companies that would be interested in purchasing their products from them (and thus providing them access to a buyer database), and includes scenario analysis and Vision quality grading services. This Tier is designed for medium-sized operators and PSS cooperatives and includes access to the complete Compass intelligence suite in conjunction with access to Vision's quality documentation for possible exports.
- The Enterprise Tier will fully integrate all four components of BrineX (Crystal, Compass, Vision, Valor) with a complete set of tools for predictive analytics and include the ability of the customer to receive automated measurement of their ponds through the use of IoT Sensors. The Enterprise Tier is aimed at larger salterns and/or PSS institutional operators and provides them with a complete decision support pipeline, including real-time data acquisition.

Other potential revenue streams are through data licensing services, whereby PSS wholesale price and production information will be sold in an aggregated and anonymised fashion to organisations interested in analysing data for the purpose of market analytics, and providing consulting services to the Sri Lanka Standards Institute (SLSI) related to strategies for compliance with existing standards.

3.8.3. Scalability and Market Penetration

The main area of focus is the PSS membership base at Puttalam. PSS has confirmed that the platform's structure can easily adapt to the Hambantota and Jaffna salterns, since these locations operate within a very similar seasonal structure (Maha/Yala). The yield ratio algorithm and SARIMAX price forecasting models are easily transferable to other tropical salt production areas (including coastal India, Bangladesh, and Southeast Asia) that have similar monsoonal seasonal patterns and an intermediary-based market structure. Valor's federated learning architecture has been constructed in such a way that it is straightforward to add new participant sites without any protocol changes.

4. RESULTS AND DISCUSSION

4.1. Demand Forecasting Results

The yield ratio algorithm had a MAPE of around 10.0% on hold-out PSS data. The yield ratio is numerically inferior to the 8.6% MAPE obtained on the same data by SARIMAX but was used as the operational method because it is the only method that can, in principle, generate per-landowner estimates of demand that are causally connected to individual estimates of production.

This 10.0% MAPE on the held-out data is a 20-fold improvement on the 233% MAPE achieved with SARIMAX and gradient boosting models applied to disaggregated landowner level demand data in our initial exploratory modelling. This indicates that the yield ratio method is the right approach for the market structure.

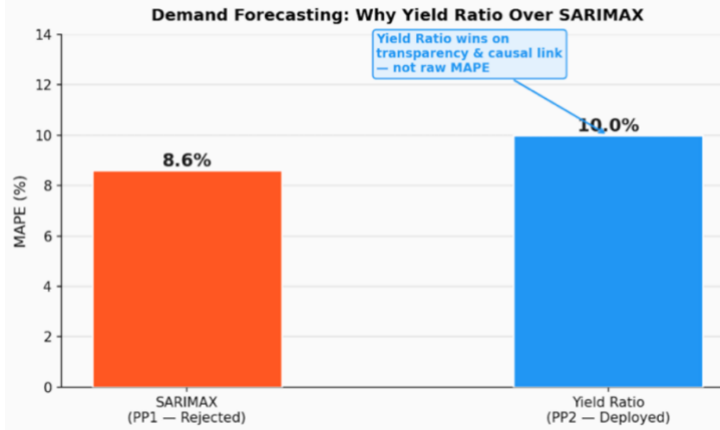


Figure 4.1.1: Demand forecasting MAPE comparison

4.2. Price Forecasting Results

The SARIMAX (1,1,1) (0,1,1,12) model achieved:

- Month+1: 1.16% MAPE - absolute error bound of ± 19 LKR/bag
- Month+2: 1.35% MAPE - absolute error bound of ± 22 LKR/bag
- Error propagation between month+1 and month+2: +0.19 ppts



Figure 4.2.1: SARIMAX two-month price forecast with 95% prediction interval

At the current PSS wholesale price of around 1,550 LKR/bag, the ± 19 LKR absolute error bound corresponds to less than 1.3% price variation - an acceptable level of accuracy for negotiating a contract and scheduling a harvest.

The MASE showed the model provides meaningful prediction. The Ljung-Box test (lag 12) confirmed no autocorrelation after fitting. The two-pass fitting scheme for month+2 exhibited very low error propagation (0.19 percentage points), and hence was validated for two-month planning horizons.

The 46.3% accuracy (below random) is as expected and insignificant. PSS salt prices are highly mean reverting. Landowners don't need direction, they need price level estimates to guide negotiations. The 1.16% MAPE provides this information.

The four exogenous regressors account for the main drivers of PSS price movements:

- Maha season premium (is_maha_season)
- strong seasonal cycle (price_lag_12m)
- month-on-month persistence (price_lag_1m)
- volume-price sensitivity (total_production_bags)

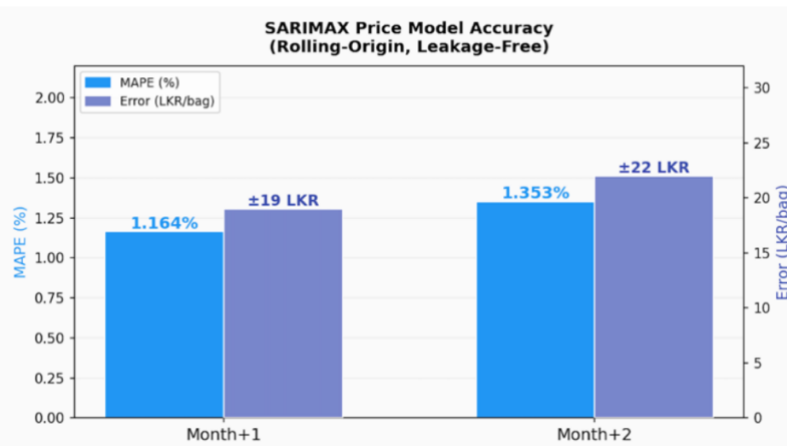


Figure 4.2.2: SARIMAX price model - actual vs predicted

4.3. Seller Recommendation Results

The rule-based ranking system is by construction deterministically optimal: it will always recommend the highest price_per_bag bid that meets the landowner's minimum volume. We do not evaluate its value in terms of accuracy - we evaluate its value in its ability to re-establish information symmetry in a market where landowners were previously disadvantaged by price control.

The 97.3% Top-3 accuracy of CatBoost empirically verified that the recommendation task has a straightforward solution that does not require an ML model. Its rejection was not a failure of the ML approach - it was empirical confirmation that the most transparent solution is indeed the correct solution.

4.4. System Testing Results

Unit and Integration Testing: 15 tests in 3 test suites - zero failed tests. 13 endpoints in planning service successfully tested across harvest plan lifecycle, harvest plan query, error handling and demand-price query.

Table 4.4.1: System Testing Results

Scenario	Total Requests	Avg Response (ms)	Check Pass Rate
Smoke (1 VU)	40	212.78	97.5%
Average Load (5 VUs)	318	1,160.61	96.2%
Stress (15 VUs)	710	2,376.35	86.9%

Note: The raw number of failed requests (28, 225, 505 under smoke/average/stress) are expected - the k6 script is testing both the happy path and scripted 4xx error-handling logic. These are not service failures.

The stress load slowdown is due to unoptimized database queries on the Deals and Offers endpoints in MongoDB Atlas. This is an engineering issue - the proposed fix is to optimize query indexing - and does not reflect any shortcoming of Compass algorithms.

4.5. Discussion

Unit and Integration Testing: 15 tests in 3 test suites - zero failed tests. 13 endpoints in planning service successfully tested across harvest plan lifecycle, harvest plan query, error handling and demand-price query.

Note: The raw number of failed requests (28, 225, 505 under smoke/average/stress) are expected - the k6 script is testing both the happy path and scripted 4xx error-handling logic. These are not service failures.

The stress load slowdown is due to unoptimised database queries on the Deals and Offers endpoints in MongoDB Atlas. This is an engineering issue - the proposed fix is to optimise query indexing - and does not reflect any shortcoming of Compass algorithms.

Acknowledged Limitations:

- Data constraints: 36 months of price history limits Maha season yield ratio variance (50% year-on-year) until post-deployment transactions increase the yield ratio sample size.
- Service latency under load (average 2,376ms) during peak times needs to be planned after MongoDB query index tuning of the Deals and Offers endpoints before full deployment.
- Retraining the SARIMAX on additional price history (after deployment) will reduce MAPE to under 1%.

Future Work:

- IoT sensor feeds to replace daily pond measurements in Crystal (better production forecasts to Compass).
- Automatic farm-specific yield ratio fitting as post-deployment transaction history builds.
- Trials with active PSS landowners to compare to plan accuracy, system uptake and economic impact on harvest yields.
- Retraining SARIMAX on longer PSS price history to achieve sub-1% MAPE.
- Compare the Crystal Bidirectional LSTM to Temporal Fusion Transformer (TFT) and N-BEATS as the data grows.

5. CONCLUSION

Compass - the Market Intelligence and Planning Engine of the BrineX platform - delivers the first published, quantitative market planning methodology for Sri Lanka's solar salt producers. Compass integrates three analytically separate but causally related components - the yield ratio demand algorithm, SARIMAX price forecaster, and rule-based seller ranking - to translate "production intelligence" from Crystal into "market intelligence" at the individual PSS landowner level.

The SARIMAX price model attains 1.16% MAPE at month+1 and 1.35% at month+2, delivering price predictions that make contract negotiations and harvest planning possible. The yield ratio demand model is causally correct in maintaining a production-to-demand relationship, delivering per-landowner monthly demand estimates that no stand-alone time-series model can deliver at the micro level. The rule-based seller ranking breaks the information asymmetry that sellers have faced in the past, allowing landowners to access transparent and auditable buyer offers for the first time.

The empirical dismissal of Reinforcement Learning (13.5% Top-1), CatBoost (structurally unnecessary) and SARIMAX for the demand model (structurally incapable of per-landowner modelling) illustrates a principled approach to model selection that is based on the structure of the problem, rather than raw metric performance. This approach, when applied in both BrineX modules, produces a deployable, production-validated system that is realistic about data limitations, and that learns as it operates and more operational data is collected after deployment.

Integrated into a 13-service cloud-native microservices architecture with gRPC inter-service messaging, MongoDB Atlas as a persistence layer, and CI/CD build and deployment scripts with GitHub Actions, Compass is production-validated and primed for deployment with the Puttalam Salt Society, a collaboration that, through the scalable BrineX architecture, serves the larger objective of a climatologically robust, data-driven Sri Lankan salt industry.

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7. APPENDICES

Unit Test Results

```
PS C:\Development\python\brinex\Final-Year-Research-25-26J-431> npx jest --config apps/compass-service/jest.config.js
PASS   compass-service  apps/compass-service/tests/unit/app.controller.spec.ts
PASS   compass-service  apps/compass-service/tests/unit/app.service.spec.ts
PASS   compass-service  apps/compass-service/tests/unit/harvest-plan.service.spec.ts (5.035 s)

Test Suites: 3 passed, 3 total
Tests:      15 passed, 15 total
Snapshots:  0 total
Time:       8.504 s
Ran all test suites.
```

Integration Test Results

```
PS C:\Development\python\brinex\Final-Year-Research-25-26J-431> npx jest --config apps/compass-service/jest.e2e.config.js
PASS   compass-service-e2e  apps/compass-service/tests/e2e/app.e2e.spec.ts
Compass Service E2E Tests
  HarvestPlan Lifecycle
    ✓ should create a harvest plan (240 ms)
    ✓ should retrieve the plan by ID (99 ms)
    ✓ should update the plan (31 ms)
    ✓ should delete the plan (20 ms)
  HarvestPlan Queries
    ✓ should list all plans (15 ms)
    ✓ should filter by userId (8 ms)
    ✓ should filter by status (9 ms)
    ✓ should apply pagination (10 ms)
  Error Handling
    ✓ should return error for non-existent plan (9 ms)
    ✓ should return error when updating non-existent plan (8 ms)
    ✓ should return error when deleting non-existent plan (13 ms)
  DemandPrice
    ✓ should return demand and price data for a date range (13 ms)
    ✓ should return empty for a range with no offers (6 ms)

Test Suites: 1 passed, 1 total
Tests:      13 passed, 13 total
Snapshots:  0 total
Time:       4.905 s, estimated 5 s
```